

A self-consistency reading of the de Vries electroweak relation

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The on-shell weak mixing angle is reproduced parameter-free by the positive root X_+ of $X^2 + CX - C = 0$ with $C = j(j+1)$ at $j_W = \frac{1}{2}$, $j_Z = 1$, to 0.42σ . We note that this root is the fixed point of the self-consistency relation $X = C/(X+C)$, a self-energy (gap-equation) structure rather than a Laplacian eigenvalue, and examine whether a unitary, positive-definite dynamical mechanism outputs it.

I. THE RELATION

The on-shell weak mixing angle and W/Z mass ratio are reproduced, with no free parameter, by the positive root of

$$X^2 + CX - C = 0, \quad X \equiv \frac{M_V^2}{\mu^2}, \quad C \equiv j(j+1), \quad (1)$$

evaluated at the spins $j_W = \frac{1}{2}$ ($C = \frac{3}{4}$) and $j_Z = 1$ ($C = 2$). With $X_+(C) = \frac{1}{2}(-C + \sqrt{C^2 + 4C})$ one has $X_+(\frac{3}{4}) = 0.56873$, $X_+(2) = 0.73205$, and

$$\sin^2 \theta_W = 1 - \frac{X_+(\frac{3}{4})}{X_+(2)} = 0.223101, \quad (2)$$

against the measured on-shell value $0.22321(26)$, a 0.42σ agreement.

II. SELF-CONSISTENCY FORM

Equation (1) is the fixed-point condition

$$X = \frac{C}{X+C}, \quad (3)$$

i.e. a field equal to a coupling times its own propagator evaluated at the self-consistent mass. The map $X \mapsto C/(X+C)$ has the stable fixed point X_+ and the spurious unstable one $X_- = -C$. The spectrum is bounded, $0 < X_+ < 1$ with $X_+ \rightarrow 1$ as $C \rightarrow \infty$: a saturating self-energy, not an additive tower $X = X_0 + n$. This places (1) in the resolvent/gap-equation class, not the quadratic-form (KK Laplacian) class.

III. A ONE-CASIMIR RESOLVENT GAP

For a massive vector in the spin- j slot of an internal $SU(2)$, with quadratic Casimir $\mathbf{T}^2 = C\mathbb{K}$, $C = j(j+1)$, on the irrep, take the order-parameter effective potential

$$\mathcal{V}_j(X) = \frac{1}{2}X^2 - C \log(X + C), \quad X \equiv \frac{M_V^2}{\mu^2}. \quad (4)$$

Its stationarity $\mathcal{V}'_j(X) = X - C/(X + C) = 0$ is the gap equation

$$X = \frac{C}{X + C} \iff \Sigma_j(X) = \mu^2 \frac{C}{X + C}, \quad \Gamma(p^2) = p^2 - \Sigma, \quad (5)$$

identical to Eq. (3). The Casimir enters once as the numerator (the vertex trace of $\mathbf{T}^2$) and once as the resolvent shift in the denominator (the dressed internal mode of self-mass $\mu^2 C$), both from the single operator $\mathbf{T}^2 = C\mathbb{K}$.

A unitary Hilbert-space realization is a rank-one Feshbach/Friedrichs two-channel Hamiltonian: an open vector channel P with $H_{PP} = 0$, a closed rotor channel Q with $H_{QQ} = \mu^2 C$, and transition $H_{PQ} = \mu^2 T^a$. The Feshbach effective Hamiltonian $H_{\text{eff}}(E) = H_{PQ}(E - H_{QQ})^{-1}H_{QP} = \mu^4 C/(E - \mu^2 C)$ has its pole at $E = -M_V^2$ where Eq. (5) holds. The pole residue is positive, $Z^{-1} = 1 + \mu^4 C/(M_V^2 + \mu^2 C)^2 > 0$, and the self-energy is a positive Stieltjes (Källén–Lehmann) form with single-line spectral density $\rho_j(\nu) = \mu^4 C \delta(\nu - \mu^2 C) \geq 0$. No ghost, no tachyon, $0 < X_+ < 1$.

Assigning the photon to the singlet $j = 0$ ($C = 0 \Rightarrow X = 0$, massless), the $W^\pm$ to the lowest center-odd channel $j = \frac{1}{2}$ ($C = \frac{3}{4}$), and the Z to the lowest center-even nonsinglet $j = 1$ ($C = 2$, from $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$), Eqs. (1)–(2) follow as outputs: $X_+(\frac{3}{4}) = (\sqrt{57} - 3)/8$, $X_+(2) = \sqrt{3} - 1$, $\sin^2 \theta_W = 0.223101$. Among half-integer pairs (j_W, j_Z) with $j \leq 4$, this lowest-spin assignment is the only one reproducing the measured angle within 3σ .